Lab2:Assignment

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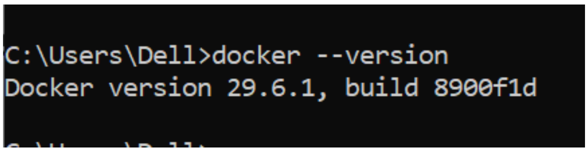
Docker - Container Management

Lab 2 : Docker Container Lifecycle and Management Using Nginx

Step 1: Check Docker Installation

docker –version



What is happening?

When you type:

docker --version

Docker checks whether the **Docker CLI (Command Line Interface)** is installed on your computer.

You
│
│ docker --version
▼
Docker CLI
│
▼
Displays Version

Example Output

Docker version 28.2.2, build 123456

This command **does not contact Docker Hub or create anything**. It simply checks the Docker program installed on your machine.

Step 2: Search for Nginx

docker search nginx

```
C:\Users\Dell>docker search nginx
NAME                                     DESCRIPTION                                     STARS     OFFICIAL
nginx                                    Official build of Nginx.                        21337     [OK]
nginx/nginx-ingress                      NGINX and  NGINX Plus Ingress Controllers fo…   121
nginx/nginx-prometheus-exporter          NGINX Prometheus Exporter for NGINX and NGIN…   52
nginx/nginx-ingress-operator             NGINX Ingress Operator for NGINX and NGINX P…   4
nginx/nginxaas-loadbalancer-kubernetes                                                   1
bitnamicharts/nginx                      Bitnami Helm chart for NGINX Open Source        4
ubuntu/nginx                             Nginx, a high-performance reverse proxy & we…   141
kasmweb/nginx                            An Nginx image based off nginx:alpine and in…   9
rancher/nginx                                                                            4
linuxserver/nginx                        An Nginx container, brought to you by LinuxS…   236
dtagdevsec/nginx                         T-Pot Nginx                                     0
paketobuildpacks/nginx                                                                   0
vmware/nginx                                                                             3
gluufederation/nginx                      A customized NGINX image containing a consu…   1
cleanstart/nginx                         Secure by Design, Built for Speed, Hardened …   0
antrea/nginx                             Nginx server used for Antrea e2e testing        0
activestate/nginx                        ActiveState's customizable, low-to-no vulner…   0
intel/nginx                                                                              0
docksal/nginx                            Nginx service image for Docksal                 1
geokrety/nginx                           Our customized nginx image                      0
circleci/nginx                           This image is for internal use                  2
ilios/nginx                              Nginx customized to run Ilios along with the…   0
corpusops/nginx                          https://github.com/corpusops/docker-images/     1
wayofdev/nginx                           Nginx Docker image for PHP development. SSL-…   0
dockette/nginx                           Nginx SSL / HSTS / HTTP2                        3
```

What happens?

Docker connects to **Docker Hub**, which is the official online repository for Docker images. Your

```
Computer
     |
     |  Search "nginx"
     ▼
Docker Hub
     |
     ▼
Returns Matching Images
```

Example Output

```
NAME              DESCRIPTION
nginx             Official build of Nginx
nginxinc/nginx
bitnami/nginx
```

Why?

Docker images are stored online. Before downloading one, you can search for available images.

Think of Docker Hub like the **Google Play Store** or **Apple App Store**, but instead of apps, it stores Docker images.

Step 3: Download Nginx

docker pull nginx

```
C:\Users\Dell>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
45ef33eb91c8: Pull complete
6a24a6ebcf30: Pull complete
0b19be717994: Pull complete
8d2092eabbc5: Pull complete
3b9f3509a826: Pull complete
062e450697fa: Pull complete
39694cde734e: Pull complete
cd64a2895680: Download complete
29456554d205: Download complete
Digest: sha256:c90e4ea5905ee69515d84cd5520e11d204df06cce71a312666cebbc4b9b68267
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quick
view nginx
```

What happens?

Docker downloads the **Nginx image** from Docker Hub to your computer.

Docker Hub
│
│ Download Image
▼
Your Computer
│
▼
Stored in Docker Images

After downloading:

docker images

shows

REPOSITORY TAG
nginx latest

Important

At this point:

+ No container exists.

Only the **image** exists.

Think of it like:

Image = Software Installer

Container = Installed & Running Software

Step 4: View Images

docker images

```
C:\Users\Dell>docker images
                                                                         i Info →   U  In Use
IMAGE                   ID             DISK USAGE   CONTENT SIZE   EXTRA
mongo-express:latest    1b23d7976f02        287MB         59.8MB    U
mongo:7                 d5b3ca8c3f3c       1.19GB          297MB    U
mongo:latest            b8806ee82073        1.3GB          339MB    U
mysql:latest            ad88e1c86cbf       1.28GB          288MB    U
nginx:latest            c90e4ea5905e        241MB           66MB
nginxapp:latest         4cabd87e1732       92.5MB           26MB    U
simpleapp:v1            53745f515b8c        203MB         49.7MB    U
```

What happens?

Docker shows all downloaded images. Example

IMAGE ID
nginx
ubuntu
mysql

Nothing is running yet. You

only have installers.

Step 5: Run Nginx

docker run nginx

```
C:\Users\Dell>docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform c
onfiguration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.s
h
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/def
ault.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/d
efault.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 08:59:22 [notice] 1#1: using the "epoll" event method
2026/07/14 08:59:22 [notice] 1#1: nginx/1.31.2
2026/07/14 08:59:22 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 08:59:22 [notice] 1#1: OS: Linux 6.6.87.1-microsoft-standard-WSL2
2026/07/14 08:59:22 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 08:59:22 [notice] 1#1: start worker processes
2026/07/14 08:59:22 [notice] 1#1: start worker process 29
2026/07/14 08:59:22 [notice] 1#1: start worker process 30
2026/07/14 08:59:22 [notice] 1#1: start worker process 31
2026/07/14 08:59:22 [notice] 1#1: start worker process 32
```

This is one of the most important Docker commands.

What happens internally?

Docker performs several actions:

Step A

Checks whether the image exists. Is

nginx image available?

YES

If not:

Docker automatically performs:

docker pull nginx

Step B

Creates a **new container** from the image.

Image

↓

New Container

The image remains unchanged.

Containers are copies of the image.

Step C

Starts the Nginx server.

Container

↓

Nginx Starts

Now Nginx is running inside the container.

Step D

Your terminal becomes attached to the running container.

Terminal

↓

Container Output

Stopping the program using Ctrl

+ C

stops the container.

Step 6: Detached Mode

docker run -d nginx

```
C:\Users\Dell>docker run -d nginx
a9bb916785a6d15bb6730157cddcab2712eec5dfc66b26ceac6ac102bf0aa348
```

What changed?

-d

means

Detached Mode

Instead of connecting your terminal to the container: Terminal

↓

Gets Prompt Back

The container continues running in the background. Docker

returns

9f3ab2c89c8d...

This is the **Container ID**.

Step 7: Name the Container

docker run -d --name mynginx nginx

```
C:\Users\Dell>docker run -d --name mynginx nginx

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container e
rror"
docker: Error response from daemon: Conflict. The container name "/mynginx" is alrea
dy in use by container "75b1c192a6ecd39dc23374b1b664e93df30afec64a4c5dff5d2973165e4e
39da". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
```

Without a name:

Docker generates random names.

Example

adoring_pike

crazy_turing

happy_einstein

Using

--name mynginx

makes management easier.

Instead of

docker stop 98ab123...

you simply type docker

stop mynginx

Step 8: Publish a Port

docker run -d --name webserver -p 8080:80 nginx

```
C:\Users\Dell>docker run -d --name webserver -p 8080:80 nginx

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container e
rror"
docker: Error response from daemon: Conflict. The container name "/webserver" is alr
eady in use by container "1c90eae4bc61fdf6d6d1de7b27fed7f8aa0a6d6b8ae1f95d9d6b36bf9f
450226". You have to remove (or rename) that container to be able to reuse that name
.

Run 'docker run --help' for more information
```

This is the most important networking concept.

Inside the container

Nginx always listens on Port

80
Container

Nginx

↓ 80

But your computer cannot automatically access that internal port. So

Docker creates a mapping.

Computer

8080

↓

Container

80

Meaning

localhost:8080

↓

Container Port 80

Opening

http://localhost:8080

actually forwards the request to

Container

↓ Nginx

↓

Port 80

Step 9: List Running Containers

docker ps

```
C:\Users\Dell>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS          PORTS                                         NAMES
a9bb916785a6   nginx     "/docker-entrypoint.…"   4 minutes ago   Up 4 minutes    80/tcp                                        amazing_brown
5f13b00b0abf   mongo:7   "docker-entrypoint.s…"   3 days ago      Up 11 minutes   0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp   college-mongodb
```

Shows

Container ID

Image Status

Ports

Name

Example

mynginx

Up 5 minutes

0.0.0.0:8080->80

Meaning

Host Port

8080

↓

Container Port

80

Step 10: Stop Container

docker stop mynginx

```
C:\Users\Dell>docker stop mynginx
mynginx
```

Docker sends a signal to Nginx. Please

Stop Gracefully

Nginx saves its state and exits cleanly.

Container Status

Running

↓Exited

Notice:

The **container is not deleted**.

It is only stopped.

Step 11: Start Container Again

docker start mynginx

```
C:\Users\Dell>docker start mynginx
mynginx
```

Docker does **not create a new container**. It

simply starts the **existing** one.

Stopped Container

↓

Running Container

Step 12: Restart

docker restart mynginx

```
C:\Users\Dell>docker restart mynginx
mynginx
```

Equivalent to

docker stop mynginx docker

start mynginx

Step 13: View Logs

docker logs mynginx

```
C:\Users\Dell>docker logs mynginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/11 06:15:07 [notice] 1#1: using the "epoll" event method
2026/07/11 06:15:07 [notice] 1#1: nginx/1.31.2
2026/07/11 06:15:07 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/11 06:15:07 [notice] 1#1: OS: Linux 6.6.87.1-microsoft-standard-WSL2
2026/07/11 06:15:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/11 06:15:07 [notice] 1#1: start worker processes
2026/07/11 06:15:07 [notice] 1#1: start worker process 29
2026/07/11 06:15:07 [notice] 1#1: start worker process 30
2026/07/11 06:15:07 [notice] 1#1: start worker process 31
2026/07/11 06:15:07 [notice] 1#1: start worker process 32
2026/07/11 06:23:38 [notice] 1#1: signal 3 (SIGQUIT) received, shutting down
2026/07/11 06:23:38 [notice] 29#29: gracefully shutting down
2026/07/11 06:23:38 [notice] 29#29: exiting
2026/07/11 06:23:38 [notice] 30#30: gracefully shutting down
2026/07/11 06:23:38 [notice] 30#30: exiting
2026/07/11 06:23:38 [notice] 29#29: exit
2026/07/11 06:23:38 [notice] 30#30: exit
2026/07/11 06:23:38 [notice] 31#31: gracefully shutting down
2026/07/11 06:23:38 [notice] 31#31: exiting
2026/07/11 06:23:38 [notice] 31#31: exit
2026/07/11 06:23:38 [notice] 32#32: gracefully shutting down
```

Every application writes messages.

Example

Nginx Started

Request Received

Response Sent

Docker stores these logs.

Useful for debugging.

Step 14: Enter the Container

docker exec -it mynginx bash

```
C:\Users\Dell>
C:\Users\Dell>docker exec -it mynginx bash
root@75b1c192a6ec:/#
```

This command opens a shell **inside the running container**. Your

Computer

↓

Docker

↓

Container

↓

Linux Bash

Now commands like

pwd
ls
cd /usr/share/nginx/html

are executed **inside the container**, not on your Windows machine. This is

very useful for troubleshooting or inspecting files.

Step 15: Remove the Container

docker rm mynginx

```
C:\Users\Dell>docker rm mynginx
mynginx
```

Docker deletes the **container only**.

Container

+ Deleted

The image is still available.

You can create a new container anytime using:

docker run nginx

Step 16: Remove the Image

docker rmi nginx

```
C:\Users\Dell>docker rmi nginx
Error response from daemon: conflict: unable to delete nginx:latest (must be forced) - container 353c4
ts referenced image c90e4ea5905e
```

Now Docker removes the **downloaded image** from your local system.

Image

+ Deleted

If any container still depends on that image, Docker will refuse to remove it until those containers are deleted.